Hands-On Lab (HOL)

Building Docker Images using Dockerfile

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1. Learning Objectives

By the end of this lab, participants will:

- Understand what a Dockerfile is
- Understand why Dockerfiles are used in DevOps workflows
- Learn common Dockerfile instructions
- Build Docker images using Dockerfile
- Run containers from custom images

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2. Learning Outcomes

After completing this lab, participants will be able to:

- Write a basic Dockerfile
- Build Docker images using the docker build command
- Understand Dockerfile instructions like FROM, RUN, CMD, EXPOSE
- Deploy a containerized web application

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3. What is a Dockerfile?

Before we start the lab, let's understand an important concept.

A **Dockerfile** is a text file that contains a set of instructions used to automatically build a Docker image.

Instead of manually configuring containers and then committing them, Dockerfiles allow us to define the entire environment as code.

This means we can:

- automate image creation

- maintain version control
- ensure consistent environments

A Dockerfile works like a **recipe**.

Example analogy:

Recipe → Dockerfile
Cooked Dish → Docker Image
Serving Plate → Docker Container

Whenever we run the **docker build** command, Docker reads the Dockerfile step by step and creates image layers.

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4. Why Do We Use Dockerfile?

Dockerfiles are used because they make image creation:

- Automated
- Reproducible
- Version controlled
- Easy to share

In real DevOps pipelines, images are almost always created using Dockerfiles instead of docker commit.

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5. Common Dockerfile Instructions

Some commonly used instructions are:

FROM
Defines the base image

RUN
Executes commands while building the image

COPY / ADD
Copies files into the image

EXPOSE
Defines which port the container will use

CMD
Defines the default command when container starts

MAINTAINER
Specifies the author of the image

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6. Lab Scenario

In this lab we will:

1. Create a Dockerfile
2. Build a custom image
3. Run a container from the image
4. Modify the Dockerfile to deploy a web application
5. Access the application using port mapping

Step 1: Check Existing Images

```
docker images
```

Step 2: Navigate to Home Directory

```
cd
```

Step 3: List Files

```
ls
```

Step 4: Create a Dockerfile

```
vi Dockerfile
```

Add the following content:

```
FROM ubuntu
RUN apt update -y
RUN apt install git -y
RUN mkdir devops jenkins
RUN touch file1
RUN echo "this is first dockerfile" >file1
```

Save and exit the file.

Step 5: Build Docker Image

First attempt (incorrect path example):

```
docker build -t test_new_image Dockefile
```

Correct command:

```
docker build -t test_new_image .
```

Explanation:

- `-t` → tag name
- `test_new_image` → image name
- `.` → build context (current directory)

Step 6: Verify Image Creation

```
docker images
```

Step 7: Run Container from Image

```
docker run -it --name c1 test_new_image
```

Inside the container verify:

```
ls
cat file1
```

Exit the container.

Step 8: Modify Dockerfile for Web Application

Edit Dockerfile again:

```
vi Dockerfile
```

Replace content with:

```
FROM ubuntu:18.04
MAINTAINER sandeep
RUN apt-get update -y
RUN apt-get install apache2 -y
RUN apt-get install apache2-utils -y
RUN echo "This is Web application Running inside Container" >/var/www/html/
index.html
EXPOSE 80
CMD ["apache2ctl","-D","FOREGROUND"]
```

Step 9: Build Web Application Image

```
docker build -t app_image .
```

Step 10: Verify Image

```
docker images
```

Step 11: Run Web Application Container

```
docker run -d --name webapplication -p 8080:80 app_image
```

Explanation:

- `-d` → detached mode
- `-p 8080:80` → port mapping

Step 12: Verify Container

```
docker ps -a
```

Step 13: Check Command History

```
history
```

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7. Test the Web Application

Open browser and access:

```
http://localhost:8080
```

You should see:

This is Web application Running inside Container

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8. Lab Summary

In this lab we learned:

- What a Dockerfile is

- Why Dockerfiles are used in DevOps
- How to write Dockerfile instructions
- How to build Docker images
- How to deploy a web application inside a container

Dockerfiles are the **industry standard way of building container images** and are widely used in CI/CD pipelines.